

Assignment 4

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Packages

```
library(tidyverse)    # Graphs with ggplot()
library(knitr)         # Tables with kable()
library(ggokabeito)   # colorblind friendly colors
library(viridis)       # colorblind friendly colors
library(brms)          # Hierarchical Bayesian models
library(rstan)         # Interface to Stan
library(shinystan)     # Model output graphs
library(bayesplot)     # Model output graphs

# General settings
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)
options(contrasts = c("contr.equalprior_pairs", "contr.poly"))
set.seed(1234) # For reproducibility
```

Preparation

Data: penguins

```
head(penguins)
```

```
##   species   island bill_len bill_dep flipper_len body_mass   sex year
## 1  Adelie Torgersen   39.1    18.7        181      3750  male 2007
## 2  Adelie Torgersen   39.5    17.4        186      3800 female 2007
## 3  Adelie Torgersen   40.3    18.0        195      3250 female 2007
## 4  Adelie Torgersen    NA      NA         NA        NA   <NA> 2007
## 5  Adelie Torgersen   36.7    19.3        193      3450 female 2007
## 6  Adelie Torgersen   39.3    20.6        190      3650  male 2007
```

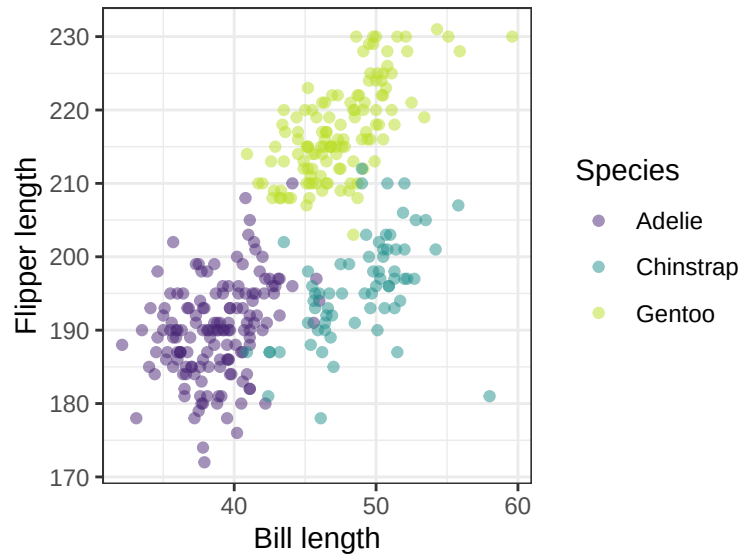
The data consist of several measurements of penguins of the Palmer archipel in Antarctica.

Variable	Description
species	Factor: Adelie, Chinstrap, Gentoo
island	Factor: Biscoe, Dream, Torgersen
bill_len	Numeric: Mean = 43.91, range [32.10, 59.6]
bill_dep	Numeric: Mean = 17.15, range [13.10, 21.5]
flipper_len	Numeric: Mean = 200.9, range [172, 231]
body_mass	Numeric: Mean = 4202, range [2700, 6300]
sex	Factor: female, male
year	Numeric: 2007, 2008, 2009

```

penguins <- penguins
g1 <- ggplot(penguins, aes(x = bill_len, y = flipper_len, color = species))+
  geom_point(alpha = .5)+
  scale_color_viridis_d(begin = .1, end = .9)+
  labs(x = "Bill length", y = "Flipper length", color = "Species")+
  theme_bw()
g1

```



```

# Remove missing values
penguins <- na.omit(penguins)

# Center numeric predictors in case you build models with interaction terms
penguins$bill_len_c <- penguins$bill_len - mean(penguins$bill_len)
penguins$bill_dep_c <- penguins$bill_dep - mean(penguins$bill_dep)

```

Exercices

Analyse the penguins dataset to investigate the hypothesis that flipper length is influenced by bill length and bill depth (at this point without considering interactions). Do so following a series of steps in the Bayesian workflow.

Exercise 1.

Translate the hypothesis into a Bayesian mathematical model. What is the general structure of this model?

Exercise 2.

Take a look at the default priors for the model, and define two sets of alternative priors. Hint: You can use the `default_prior()` function to see the default prior structure for this model.

Exercise 3.

Conduct a small sensitivity analysis by comparing the prior predictive values of the model for the three sets of priors.

Exercise 4.

Apply the model to the data with `brm()`, using the priors that faired best in the prior predictive checks. Assess the convergence parameters and briefly interpret the model output.

Exercise 5.

Define two alternative models: An intercept-only model without predictors, and a complexer model that includes not just the main effects of `bill_len` and `bill_dep`, but also their interaction. Compare these models using Bayes factors, and briefly interpret the results.

Exercise 6

Conduct posterior predictive checks for the model that fairs best on the basis of the Bayes factor comparison.